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Supplemental Information

Catalytic Coupling of Oxidative Phosphorylation, ATP Demand, and Reactive Oxygen Species Generation

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Supporting Materials for "Catalytic Coupling of Oxidative Phosphorylation, ATP Demand, and Mitochondrial Reactive Oxygen Species Generation"

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The Supporting Materials presents a detailed description of the integrated model used to simulate the results in the main paper. For convenience, the model code is included as supplemental material.

Integrated Model of Mitochondrial Bioenergetics and ROS Dynamics

The present model is extended from the previous model of mitochondrial bioenergetics [1]. Several flux expressions and model parameters were updated based on the recent works of Bazil et al. [2, 3], as laid out in this model supplemental material section. This supplement consists of four parts. Part 1 lists all the state variables comprising the model, dissociation constants, and physiochemical and general model parameters. Part 2 introduces the set of fifteen non-linear ODEs, five algebraic conservation expressions (for mitochondrial ATP, NADH, UQH₂, inter-membrane space c²⁺, and extra-mitochondrial creatine phosphate), and six non-linear cation ODEs (mitochondrial and extra-mitochondrial H⁺, K⁺, and Mg²⁺). Equations 2.11, 2.13, and 2.20 were only used for the ischemia/reperfusion studies detailed in the main article and in section S3C. Part 3 presents the model rate equations and the associated parameter definitions and values. Part 4 details the simulation protocols used to generate the figures, sensitivity analysis, and control coefficient calculations. Part 5 discusses a steady state analysis of the model. Part 6 describes the code used to simulate the model and generate the plots given in the paper.

The outer-mitochondrial membrane (OMM) is highly permeable to ions, metabolites and substrates of low molecular weight under the given conditions. As such, all permeable inter-membrane space (IMS) state variables were replaced with their respective extra-mitochondrial counterparts, and thus not included in the model. This has the benefit of making the system of non-linear ODEs less stiff and shortens simulation time without significantly altering the simulation results. The only IMS state variables explicitly simulated in the model are oxidized and reduced forms of cytochrome c (c³⁺ and c²⁺).

S1 - Model State Variables, Dissociation Constants, and Physiochemical and General Parameter Values

The state variables used in the model, their definitions, and their units are provided in Table S1.1. All the dissociation constants used in the model are presented in Table S1.2. Most of the dissociation constants are taken from Li et al. [4] and corrected for appropriate temperature (T = 37 °C) and ionic strength (I = 0.17 M), corresponding to the present experiments. The physiochemical and general model parameters are presented in Table S1.3.

Table S1.1. Model State Variables

State Variable	Definition	Units
$\Delta\Psi$	Mitochondrial membrane potential	mV
<i>Mitochondrial State Variables</i>		
$[H^+]_x$	Mitochondrial free proton concentration	M
$[K^+]_x$	Mitochondrial free potassium concentration	M
$[Mg^{2+}]_x$	Mitochondrial free magnesium concentration	M
$[ATP]_x$	Total mitochondrial ATP concentration	M
$[ADP]_x$	Total mitochondrial ADP concentration	M
$[Pi]_x$	Total mitochondrial Pi concentration	M
$[NADH]_x$	Total mitochondrial NADH concentration	M
$[NAD]_x$	Total mitochondrial NAD concentration	M
$[UQH_2]_x$	Total mitochondrial ubiquinol concentration	M
$[UQ]_x$	Total mitochondrial ubiquinone concentration	M
$[O_2^-]_x$	Total mitochondrial superoxide concentration	M
$[H_2O_2]_x$	Total mitochondrial hydrogen peroxide concentration	M
<i>Intermembrane Space (IMS) State Variables</i>		
$[c^{2+}]_i$	Total IMS cytochrome c^{2+} (reduced) concentration	M
$[c^{3+}]_i$	Total IMS cytochrome c^{3+} (oxidized) concentration	M
<i>Extra-Mitochondrial State Variables</i>		
$[H^+]_e$	Extra-mitochondrial free proton concentration	M
$[K^+]_e$	Extra-mitochondrial free potassium concentration	M
$[Mg^{2+}]_e$	Extra-mitochondrial free magnesium concentration	M
$[ATP]_e$	Total extra-mitochondrial ATP concentration	M
$[ADP]_e$	Total extra-mitochondrial ADP concentration	M
$[AMP]_e$	Total extra-mitochondrial AMP concentration	M
$[Pi]_e$	Total extra-mitochondrial Pi concentration	M
$[O_2^-]_e$	Total extra-mitochondrial superoxide concentration	M
$[H_2O_2]_e$	Total extra-mitochondrial hydrogen peroxide concentration	M
$[CrP]_e$	Total extra-mitochondrial creatine phosphate concentration	M
$[Cr]_e$	Total extra-mitochondrial creatine concentration	M

Table S1.2. Dissociation Constants at 37 °C, I = 0.17 M

Parameter	Definition	Value	Reference
K_{ATP}^H	Proton ATP binding constant	$10^{-6.33}$ M	[4]
K_{ATP}^K	Potassium ATP binding constant	$10^{-1.02}$ M	[4]
K_{ATP}^{Mg}	Magnesium ATP binding constant	$10^{-3.88}$ M	[4]
K_{ADP}^H	Proton ADP binding constant	$10^{-6.26}$ M	[4]
K_{ADP}^K	Potassium ADP binding constant	$10^{-0.89}$ M	[4]
K_{ADP}^{Mg}	Magnesium ADP binding constant	$10^{-3.00}$ M	[4]
K_{Pi}^H	Proton Pi binding constant	$10^{-6.62}$ M	[4]
K_{Pi}^K	Potassium Pi binding constant	$10^{-0.42}$ M	[4]
K_{Pi}^{Mg}	Magnesium Pi binding constant	$10^{-1.66}$ M	[4]

Table S1.3. Physiochemical and General Model Parameters^a

Parameter	Definition	Value	Reference
R	Ideal gas constant	8.314×10^{-3} kJ/K/mol	Physical constant
T	Temperature	310.15 K	Physical constant
F	Faraday's constant	96.487×10^{-3} kJ/mV/mol	Physical constant
C_{mito}	IMM capacitance	6.76×10^{-6} mol/s/l mito/mV	[1]
N_{tot}	Total NAD concentration	3 mM	[1]
Q_{tot}	Total UQ concentration	20 mM	[3]
C_{tot}	Total Cyt C concentration	2.7 mM	[1]
A_{tot}	Total AdN concentration	10 mM	[1]
Cr_{tot}	Total Cr concentration	20 mM	[5]
ρ_m	Protein density of mitochondria	2.725×10^5 mg/l mito	[6]
CS_m	Citrate synthase activity per mg mitochondria	5.62 U CS / mg	-
Vol_m	Matrix H ₂ O volume fraction	0.65 l mtz/l mito	^b
Vol_i	IMS H ₂ O volume fraction	0.072 l ims/l mito	^b
Vol_e	buffer H ₂ O volume fraction	4.54×10^3 l buffer/l mito	^c
$d_f G_{H_2O_2}^0$	H ₂ O ₂ formation energy	-134.03 kJ/mol	[7]
$d_f H_{H_2O_2}^0$	H ₂ O ₂ formation enthalpy	-194.17 kJ/mol	[7]
$z_{H_2O_2}$	H ₂ O ₂ valence	0	[7]
$nH_{H_2O_2}$	Number of protons	2	[7]
$d_f G_{O_2^-}^0$	O ₂ ⁻ formation energy	13.18 kJ/mol	[8]
$d_f H_{O_2^-}^0$	O ₂ ⁻ formation enthalpy	0 kJ/mol	[8]
$z_{O_2^-}$	O ₂ ⁻ valence	-1	[8]
$nH_{O_2^-}$	Number of protons	0	[8]

Abbreviations: IMM, inner mitochondrial membrane; NAD, nicotinamide adenine dinucleotide; UQ, ubiquinone; Cyt C, cytochrome c; AdN, adenine nucleotide; Cr, creatine.

^a All thermodynamic data obtained from Li et al. [4]. Additional data is included in this table.

^b Approximate mitochondrial matrix and IMS volumes.

^c Based on a mitochondrial load of 0.67 U CS in a 2 ml chamber.

S2 - Model Differential-Algebraic Equations

S2.2A - Model Differential Equations (Bioenergetics)

Mitochondrial Membrane Potential:

$$\frac{d\Delta\Psi}{dt} = (4J_{C1} + 2J_{C3} + 4J_{C4} - nH_{F_1F_0} J_{F_1F_0} - J_{ANT} - J_{Hleak}) / C_{mito} \quad (2.1)$$

Mitochondrial State Variables:

$$\frac{d[ADP]_x}{dt} = (J_{ANT} - J_{F_1F_0}) / Vol_m \quad (2.2)$$

$$\frac{d[Pi]_x}{dt} = (J_{PIC} - J_{F_1F_0}) / Vol_m \quad (2.3)$$

$$\frac{d[NAD]_x}{dt} = (J_{C1}^{NADH} - J_{DH}) / Vol_m \quad (2.4)$$

$$\frac{d[UQ]_x}{dt} = ((2-1)J_{C3}^{UQH_2} + \alpha_{C2}J_{DH} - J_{C1}^{UQ}) / Vol_m \quad (2.5)$$

$$\frac{d[O_2^-]_x}{dt} = \left(\frac{1}{2} J_{C3}^{SO} + J_{C1}^{SO} \right) / Vol_m - 2J_{SODx} \quad (2.6)$$

$$\frac{d[H_2O_2]_x}{dt} = J_{C1}^{H_2O_2} / Vol_m + J_{SODx} - J_{H_2O_2} - J_{H_2O_2perm} \quad (2.7)$$

Inter-Membrane Space (IMS) State Variables:

$$\frac{d[c^{3+}]_i}{dt} = (2J_{C4} - J_{C3}^{c3}) / Vol_i \quad (2.8)$$

Extra-Mitochondrial State Variables:

$$\frac{d[ATP]_e}{dt} = J_{ANT} / Vol_e + J_{AK} + J_{CK} - J_{ATPase} \quad (2.9)$$

$$\frac{d[ADP]_e}{dt} = -J_{ANT} / Vol_e - 2J_{AK} + J_{ATPase} \quad (2.10)$$

$$\frac{d[AMP]_e}{dt} = J_{AK} \quad (2.11)$$

$$\frac{d[Pi]_e}{dt} = -J_{PIC} / Vol_e + J_{ATPase} \quad (2.12)$$

$$\frac{d[Cr]_e}{dt} = J_{CK} \quad (2.13)$$

$$\frac{d[O_2^-]_e}{dt} = \frac{1}{2} J_{C3}^{SO} / Vol_e - 2J_{SODe} \quad (2.14)$$

$$\frac{d[H_2O_2]_e}{dt} = J_{SOD} + J_{H_2O_2perm} Vol_m / Vol_e \quad (2.15)$$

S2.2B - Mitochondrial Conservation Algebraic Equations

The mitochondrial species for adenine nucleotides (AdNs), nicotinamide adenine dinucleotides (NADH), ubiquinone (UQH₂), reduced cytochrome c (c²⁺), and extra-mitochondrial creatine phosphate (PCr) are conserved in the model. The conservation is implemented by using an algebraic expression to govern the

conservation. This is done using a mass matrix with the integration algorithm and setting the appropriate rows equal to a zero row vector.

$$[ATP]_m = A_{tot} - [ADP]_m \quad (2.16)$$

$$[NADH]_m = N_{tot} - [NAD]_m \quad (2.17)$$

$$[UQH_2]_m = Q_{tot} - [UQ]_m \quad (2.18)$$

$$[c^{2+}]_i = C_{tot} - [c^{3+}]_i \quad (2.19)$$

$$[PCr]_e = CR_{tot} - [Cr]_e \quad (2.20)$$

S2.2C - Model Differential Equations (Cations)

The cation differential equations for the mitochondrial and extra-mitochondrial compartments are derived using the method outlined in Vinnakota et al. [9]. The general method will be presented with the understanding that compartment specific concentrations and transport rates are to be used where appropriate. In some cases, these compartment specific equations will be given. Due to the large expressions resulting from the derivation, the method used to obtain them is presented versus explicitly showing all the terms that enter the differential equations. The equations used in the model are obtained by solving the linear system of equations given in Equation 2.19.

The generalized system of equations relating the cation differential equations is

$$\begin{bmatrix} \frac{d[H^+]}{dt} \\ \frac{d[K^+]}{dt} \\ \frac{d[Mg^{2+}]}{dt} \end{bmatrix} = \begin{bmatrix} 1 + H_{\partial H} & H_{\partial K} & H_{\partial Ca} \\ K_{\partial H} & 1 + K_{\partial K} & K_{\partial Mg} \\ Mg_{\partial H} & Mg_{\partial K} & 1 + Mg_{\partial Mg} \end{bmatrix}^{-1} \begin{bmatrix} \Phi_H \\ \Phi_K \\ \Phi_{Mg} \end{bmatrix}, \quad (2.21)$$

where $X_{\partial Y}$ is the partial derivative of the concentration of *bound* X with respect to Y (Y can equal X) and Φ_X is the flux of X into or out of the compartment.

Assuming higher order cation binding is negligible, the partial derivative expressions are defined below where N_r is the number of reactants, L_i is the i^{th} ligand (such as ATP, ADP, and Pi), K_i^j is the dissociation constant for the i^{th} ligand and j^{th} cation couple and P_i is the binding polynomial for the i^{th} ligand as originally defined by Alberty [7].

$$H_{\partial H} = \frac{\partial[H_{bound}]}{\partial[H^+]} = \sum_{i=1}^{N_r} \frac{[L_i](1 + [K^+]/K_i^K + [Mg^{2+}]/K_i^{Mg})}{K_i^H P_i^2} \quad (2.22)$$

$$H_{\partial K} = \frac{\partial[H_{bound}]}{\partial[K^+]} = -[H^+] \sum_{i=1}^{N_r} \frac{[L_i]/K_i^H}{K_i^K P_i^2} \quad (2.23)$$

$$H_{\partial Mg} = \frac{\partial[H_{bound}]}{\partial[Mg^{2+}]} = -[H^+] \sum_{i=1}^{N_r} \frac{[L_i]/K_i^H}{K_i^{Mg} P_i^2} \quad (2.24)$$

$$K_{\partial H} = \frac{\partial[K_{bound}]}{\partial[H^+]} = -[K^+] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^K}{K_i^H P_i^2} \quad (2.25)$$

$$K_{\partial K} = \frac{\partial[K_{bound}]}{\partial[K^+]} = \sum_{i=1}^{Nr} \frac{[L_i](1+[H^+]/K_i^H + [Mg^{2+}]/K_i^{Mg})}{K_i^K P_i^2} \quad (2.26)$$

$$K_{\partial Mg} = \frac{\partial[K_{bound}]}{\partial[Mg^{2+}]} = -[K^+] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^K}{K_i^{Mg} P_i^2} \quad (2.27)$$

$$Mg_{\partial H} = \frac{\partial[Mg_{bound}]}{\partial[H^+]} = -[Mg^{2+}] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^{Mg}}{K_i^H P_i^2} \quad (2.28)$$

$$Mg_{\partial K} = \frac{\partial[Mg_{bound}]}{\partial[K^+]} = -[Mg^{2+}] \sum_{i=1}^{Nr} \frac{[L_i]/K_i^{Mg}}{K_i^K P_i^2} \quad (2.29)$$

$$Mg_{\partial Mg} = \frac{\partial[Mg_{bound}]}{\partial[Mg^{2+}]} = \sum_{i=1}^{Nr} \frac{[L_i](1+[H^+]/K_i^H + [K^+]/K_i^K)}{K_i^{Mg} P_i^2} \quad (2.30)$$

and

$$P_i = 1 + [H^+]/K_i^H + [K^+]/K_i^K + [Mg^{2+}]/K_i^{Mg} \quad (2.31)$$

Additional Buffering in the Mitochondrial Compartment

To account for additional buffering not attributed to metabolites and substrates in the mitochondrial compartment, equations 2.22 and 2.30 are modified by adding the following terms:

$$H_{\partial H} = H_{\partial H} + \frac{[B_H]}{K_H (1 + [H^+] / K_H)^2} \quad (2.32)$$

$$Mg_{\partial Mg} = Mg_{\partial Mg} + \frac{[B_{Mg}]}{K_{Mg} (1 + [Mg^{2+}] / K_{Mg})^2} \quad (2.33)$$

Table S2.1. Additional Mitochondrial Buffering Parameters

Parameter	Definition	Value	Reference
$[B_H]$	Total H^+ binding sites	20 mM	[10]
K_H	H^+ binding constant	$10^{-7.5}$ M	[10]
$[B_{Mg}]$	Total Mg^{2+} binding sites	8.7 mM	[11] ^a
K_{Mg}	Mg^{2+} binding constant	340 μ M	[11]

^a based on 32 nmol/mg

Flux Terms

The terms representing flux into a given compartment are defined as

$$\Phi_H = - \sum_{i=1}^{Nr} \frac{\partial[H_{bound}]}{\partial[L_i]} \frac{d[L_i]}{dt} + \sum_{k=1}^{Nk} n_k J_k + J_t^H \quad (2.34)$$

$$\Phi_K = - \sum_{i=1}^{Nr} \frac{\partial[K_{bound}]}{\partial[L_i]} \frac{d[L_i]}{dt} + J_t^K \quad (2.35)$$

and

$$\Phi_{Mg} = - \sum_{i=1}^{Nr} \frac{\partial[Mg_{bound}]}{\partial[L_i]} \frac{d[L_i]}{dt} + J_t^{Mg} \quad (2.36)$$

where N_k is the number of reactions, n_k is the stoichiometric coefficient of k^{th} reaction, J_k is the k^{th} reaction rate and J_t^i is the i^{th} cation transport rate into the compartment.

The ligand dependent partial derivative expressions are defined as

$$\frac{\partial[H_{bound}]}{\partial[L_i]} = [H^+] \sum_{i=1}^{Nr} \frac{1}{K_i^H P_i} \quad (2.37)$$

$$\frac{\partial[K_{bound}]}{\partial[L_i]} = [K^+] \sum_{i=1}^{Nr} \frac{1}{K_i^K P_i} \quad (2.38)$$

and

$$\frac{\partial[Mg_{bound}]}{\partial[L_i]} = [Mg^{2+}] \sum_{i=1}^{Nr} \frac{1}{K_i^{Mg} P_i} \quad (2.39)$$

Compartment Reactions and Fluxes

The generation of protons by biochemical reactions in the mitochondria is defined as

$$\sum_{k=1}^{Nk} n_k J_k = J_{DH} \quad (2.40)$$

This term is zero in the extra-mitochondrial compartment.

The transport of cations into the mitochondrial compartment are defined as

$$J_t^H = \left(J_{DH} + (nH_{F_1F_0} - 1) J_{F_1F_0} + 2J_{PIC} + J_{KHE} + J_{Hleak} - 5J_{C1} - 2J_{C3} - 4J_{C4} \right) \quad (2.41)$$

$$J_t^K = -J_{KHE} \quad (2.42)$$

and

$$J_t^{Mg} = 0 \quad (2.43)$$

The transport terms for the extra-mitochondrial compartment are the negative of their mitochondrial counterparts.

S3: Model Rate Equations

Each reaction rate and transport mechanism in the model is presented below. First, the electron transport related reaction rates are shown. Next, the oxidative phosphorylation related reaction rates and transport mechanisms are discussed. Finally, the cation transport and ROS scavenging mechanisms are presented.

S3A - Electron Transport Related Reaction Rates

Substrate oxidation and the electron transport system are based on the rate equations presented in Beard [1]. The mitochondrial dehydrogenase rate (substrate oxidation rate) will be introduced first, followed by the electron transport system rate equations and finally the passive proton leak rate equation is presented.

Mitochondrial Dehydrogenases

The biochemical equation for the mitochondrial dehydrogenase is defined as



The rate expression used in the model is

$$J_{DH} = \frac{X_{DH}}{1 + \left(\frac{[ATP]_x K_{DH}}{[ADP]_x [Pi]_x} \right)^{n_{DH}}} (r_{DH} [NAD^+]_x - [NADH]_x).$$

Table S3.1. Mitochondrial Dehydrogenase Parameters^b

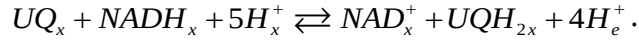
Parameter	Definition	Value	Reference
X_{DH}	Dehydrogenase activity	1.12×10^{-1} mol/lmito/s	- ^a
r_{DH}	Apparent NADH/NAD ⁺ equilibrium constant	35.4 unitless	- ^a
K_{DH}	ATPase feedback constant	2.87×10^{-2} M	- ^a
n_{DH}	ATPase Hill coefficient	1.45 unitless	- ^a

^a Adjustable parameter.

^b In the model code, the rate expression was made to saturate at $[NAD^+]_x$ above 2.58 mM.

NADH-ubiquinone oxidoreductase: Complex I

The biochemical equation for Complex I is defined as



For model details, see [2]. In brief, the model is a five-state model that includes the minimal components necessary to simulate NADH-quinone oxidoreductase activity as a function of pH and mitochondrial membrane potential ($\Delta\Psi$) and also the detailed redox biochemistry required to simulate ROS generation by both the FMN and the SQ sites. The model consists of numerous binding polynomials that represent various combinations free, substrate-bound, product-bound, and protonated states and characterizes the multiple redox states of the enzyme. In addition, the thermodynamic equations are corrected for pH, temperature, and ionic strength. The model equations consist of about a hundred supporting equations that culminate into the following matrix equation describing the steady state fractional occupancies of the differing redox states:

$$\begin{bmatrix} -(k_{0 \rightarrow 1} + k_{0 \rightarrow 2}) & k_{1 \rightarrow 0} & k_{2 \rightarrow 0} & 0 & 0 \\ k_{0 \rightarrow 1} & -(k_{1 \rightarrow 0} + k_{1 \rightarrow 2} + k_{1 \rightarrow 3}) & k_{2 \rightarrow 1} & k_{3 \rightarrow 1} & 0 \\ k_{0 \rightarrow 2} & k_{1 \rightarrow 2} & -(k_{2 \rightarrow 0} + k_{2 \rightarrow 1} + k_{2 \rightarrow 3} + k_{2 \rightarrow 4}) & k_{3 \rightarrow 2} & k_{4 \rightarrow 2} \\ 0 & k_{1 \rightarrow 3} & k_{2 \rightarrow 3} & -(k_{3 \rightarrow 1} + k_{3 \rightarrow 2} + k_{3 \rightarrow 4}) & k_{4 \rightarrow 3} \\ 0 & 0 & k_{2 \rightarrow 4} & k_{3 \rightarrow 4} & -(k_{4 \rightarrow 2} + k_{4 \rightarrow 3}) \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} S0 \\ S1 \\ S2 \\ S3 \\ S4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

For definitions of the individual rate constants and their constituents (kN 's, kSO 's, and kH 's), see the original article. After solving for the state occupancy vector, the rate of NADH oxidation is given as:

$$J_{C1}^{NADH} = E_{totC1} \left(kN_f^{0 \rightarrow 2} S0 + kN_f^{1 \rightarrow 3} S1 + (kN_f^{2 \rightarrow 4} - kN_r^{0 \rightarrow 2}) S2 - kN_r^{1 \rightarrow 3} S3 - kN_r^{2 \rightarrow 4} S4 \right).$$

The rate of superoxide generation is given as:

$$J_{C1}^{SO} = E_{totC1} \left(-kSO_r^{1 \rightarrow 0} S0 + (kSO_f^{1 \rightarrow 0} - kSO_r^{2 \rightarrow 1}) S1 + (kSO_f^{2 \rightarrow 1} - kSO_r^{3 \rightarrow 2}) S2 + (kSO_f^{3 \rightarrow 2} - kSO_r^{4 \rightarrow 3}) S3 + kSO_f^{4 \rightarrow 3} S4 \right).$$

The rate of hydrogen peroxide generation is given as:

$$J_{C1}^{H_2O_2} = E_{totC1} \left(-kH_r^{2 \rightarrow 0} S0 - kH_r^{3 \rightarrow 1} S1 + (kH_f^{2 \rightarrow 0} - kH_r^{4 \rightarrow 2}) S2 + kH_f^{3 \rightarrow 1} S3 + kH_f^{4 \rightarrow 2} S4 \right).$$

And for numerical stability, the rate of UQ reduction is coupled to the above rates such that:

$$J_{C1}^{UQ} = J_{C1}^{NADH} - J_{C1}^{SO} / 2 - J_{C1}^{H_2O_2}.$$

Below is an updated list of parameters.

Table S3.2. NADH-Ubiquinone Oxidoreductase Parameters^a

Parameter	Definition	Value	Reference
Structural Parameters			
$E_m^0(FMN / FMNH_2)$	FMN/FMNH ₂ midpoint potential	55.14 mV	[12] ^b
$E_m^0(FMN / FMNH^{\cdot-})$	FMN/FMNH ^{·-} midpoint potential	23.54 mV	[12] ^b

$E_m^0(FMNH^{\cdot-} / FMNH_2)$	FMNH ^{·-} /FMNH ₂ midpoint potential	86.74 mV	[12] ^b
pK_{FMNH_2}	FMNH ₂ pK	7.1	[12] ^b
$pK_{FMNH^{\cdot-}}$	FMNH ^{·-} pK	7.9	[12] ^b
$E_m^0(N2_{ox} / N2_{red})$	N2 _{ox} /N2 _{red} midpoint potential	-90 mV	[13] ^c
$pK_{N2_{ox}}$	N2 _{ox} pK	6	[13] ^c
$pK_{N2_{red}}$	N2 _{red} pK	8.5	[13] ^c
Kinetic Parameters			
E_{totC1}	Total Complex I content	1.67x10 ⁻⁴ mol/lmito	- ^d
K_{NADH}^{ox}	NADH dissociation constant for oxidized FMN	4.61E-05 M	[2]
$K_{NAD^+}^{ox}$	NAD ⁺ dissociation constant for oxidized FMN	7.05E-04 M	[2]
K_{NADH}^{red}	NADH dissociation constant for reduced FMN	4.99E-04 M	[2]
$K_{NAD^+}^{red}$	NAD ⁺ dissociation constant for reduced FMN	1.18E-05 M	[2]
K_{NADH}^{rad}	NADH dissociation constant for FMN radical	1.00E+00 M	[2]
$K_{NAD^+}^{rad}$	NAD ⁺ dissociation constant for FMN radical	1.54E-04 M	[2]
$K_{Q_{10}H_2}$	Q ₁₀ H ₂ dissociation constant	1.00E-01 M	[2]
$K_{Q_{10}}$	Q ₁₀ dissociation constant	1.75E-02 M	[2]
$K_{Q_{10}}^{stability}$	Q ₁₀ stability constant	1.00E+01	[2]
$k_{f,0 \rightarrow 2}^{NADH}$	NADH oxidation rate for state 0	1.18E+05 s ⁻¹	[2]
$k_{f,2 \rightarrow 4}^{NADH}$	NADH oxidation rate for state 2	1.12E+04 s ⁻¹	[2]
$k_{f,1 \rightarrow 3}^{NADH}$	NADH oxidation rate for state 1	2.77E+02 s ⁻¹	[2]
$k_{f,2 \rightarrow 0}^Q$	Q reduction rate for state 2	3.49E+05 s ⁻¹	[2]
$k_{f,4 \rightarrow 2}^Q$	Q reduction rate for state 4	4.67E+12 s ⁻¹	[2]
$k_{f,3 \rightarrow 1}^Q$	Q reduction rate for state 3	5.20E+02 s ⁻¹	[2]
β	Charge translocation parameter	5.00E-01	[2]
pK_{Nsite}	NADH oxidase site pK	7.39	[2]
pK_{Qsite}	Q reductase site pK	6.41	[2]
ROS Parameters			
$k_{f,1 \rightarrow 0}^{SQ^{\cdot-}/O_2^{\cdot-}}$	Superoxide production rate from semiquinone for state 1	1.67E+09 M ⁻¹ s ⁻¹	[2]
$k_{fa,2 \rightarrow 1}^{SQ^{\cdot-}/O_2^{\cdot-}}$	Superoxide production rate from semiquinone for state 2a	2.06E-03 M ⁻¹ s ⁻¹	[2]
$k_{fb,2 \rightarrow 1}^{SQ^{\cdot-}/O_2^{\cdot-}}$	Superoxide production rate from semiquinone for state 2b	2.07E-03 M ⁻¹ s ⁻¹	[2]
$k_{fa,3 \rightarrow 2}^{SQ^{\cdot-}/O_2^{\cdot-}}$	Superoxide production rate from semiquinone for state 3a	1.14E-05 M ⁻¹ s ⁻¹	[2]

$k_{fb,3\rightarrow 2}^{SQ^-/O_2^-}$	Superoxide production rate from semiquinone for state 3b	$2.40E-09\text{ M}^{-1}\text{s}^{-1}$	[2]
$k_{f,4\rightarrow 3}^{SQ^-/O_2^-}$	Superoxide production rate from semiquinone for state 4	$2.16E-07\text{ M}^{-1}\text{s}^{-1}$	[2]
$k_{f,2\rightarrow 1}^{FMNH_2/O_2^-}$	Superoxide production rate from FMNH ₂ for state 2	$4.52E+05\text{ M}^{-1}\text{s}^{-1}$	[2]
$k_{fa,3\rightarrow 2}^{FMNH_2/O_2^-}$	Superoxide production rate from FMNH ₂ for state 3a	$5.38E-04\text{ M}^{-1}\text{s}^{-1}$	[2]
$k_{fb,3\rightarrow 2}^{FMNH_2/O_2^-}$	Superoxide production rate from FMNH ₂ for state 3b	$2.83E-05\text{ M}^{-1}\text{s}^{-1}$	[2]
$k_{f,4\rightarrow 3}^{FMNH_2/O_2^-}$	Superoxide production rate from FMNH ₂ for state 4	$1.40E+05\text{ M}^{-1}\text{s}^{-1}$	[2]
$k_{f,1\rightarrow 0}^{FMNH^-/O_2^-}$	Superoxide production rate from FMNH ⁻ for state 1	$6.81E+07\text{ M}^{-1}\text{s}^{-1}$	[2]
$k_{fa,2\rightarrow 1}^{FMNH^-/O_2^-}$	Superoxide production rate from FMNH ⁻ for state 2a	$4.04E-11\text{ M}^{-1}\text{s}^{-1}$	[2]
$k_{fb,2\rightarrow 1}^{FMNH^-/O_2^-}$	Superoxide production rate from FMNH ⁻ for state 2b	$1.19E-07\text{ M}^{-1}\text{s}^{-1}$	[2]
$k_{f,3\rightarrow 2}^{FMNH^-/O_2^-}$	Superoxide production rate from FMNH ⁻ for state 3	$1.40E-02\text{ M}^{-1}\text{s}^{-1}$	[2]
$k_{f,2\rightarrow 0}^{FMNH_2/H_2O_2}$	Hydrogen peroxide production rate from FMNH ₂ for state 2	$7.82E-07\text{ M}^{-1}\text{s}^{-1}$	[2]
$k_{fa,3\rightarrow 1}^{FMNH_2/H_2O_2}$	Hydrogen peroxide production rate from FMNH ₂ for state 3a	$2.12E+06\text{ M}^{-1}\text{s}^{-1}$	[2]
$k_{fb,3\rightarrow 1}^{FMNH_2/H_2O_2}$	Hydrogen peroxide production rate from FMNH ₂ for state 3b	$1.76E+06\text{ M}^{-1}\text{s}^{-1}$	[2]
$k_{f,4\rightarrow 2}^{FMNH_2/H_2O_2}$	Hydrogen peroxide production rate from FMNH ₂ for state 4	$6.21E+01\text{ M}^{-1}\text{s}^{-1}$	[2]

^a Updated from [2].

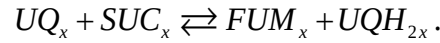
^b Refit from data in Figures 1, 2 and 3 of Sled et al. [12].

^c Refit from data in Figure 1 of Ingledew and Ohnishi [13].

^d Adjustable parameter.

Succinate-ubiquinone oxidoreductase: Complex II

The biochemical equation for Complex II is defined as



The rate expression used in the model is

$$J_{C2} = \alpha_{C2} J_{DH}.$$

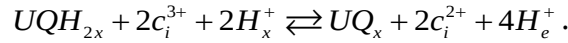
Table S3.2. Succinate-Ubiquinone Oxidoreductase Parameters

Parameter	Definition	Value	Reference
α_{C2}	Complex II activity	0.25	^a

^aThis value is bounded between 0 and 0.25 when pyruvate and malate are the substrates for oxidation. Values less than 0.25 mean incomplete oxidation of pyruvate, and there is loss of TCA cycle intermediates before reaching Complex II.

Ubiquinol-cytochrome-c oxidoreductase: Complex III

The biochemical equation for Complex III is defined as



For model details, see [3]. In brief, the model is a six-state model that simulates quinol-cytochrome c oxidoreductase activity as a function of pH and $\Delta\Psi$, as well as, the redox biochemistry involved in superoxide formation.

The model consists of numerous binding polynomials that represent various combinations free, substrate-bound, product-bound, and protonated states and characterizes the multiple redox states of the enzyme. In addition, the thermodynamic equations are corrected for pH, temperature, and ionic strength. The model equations consist of dozens of supporting equations that culminate into the following matrix equation describing the steady state fractional occupancies of the differing redox states:

$$\begin{pmatrix} -(k_{1 \rightarrow 2} + k_{1 \rightarrow 4}) & k_{2 \rightarrow 1} & 0 & k_{4 \rightarrow 1} & 0 & 0 \\ k_{1 \rightarrow 2} & -(k_{2 \rightarrow 1} + k_{2 \rightarrow 3} + k_{2 \rightarrow 5} + k_{2 \rightarrow 6}) & k_{3 \rightarrow 2} & 0 & k_{5 \rightarrow 2} & k_{6 \rightarrow 2} \\ 0 & k_{2 \rightarrow 3} & -(k_{3 \rightarrow 2} + k_{3 \rightarrow 4}) & k_{4 \rightarrow 3} & 0 & 0 \\ k_{1 \rightarrow 4} & 0 & k_{3 \rightarrow 4} & -(k_{4 \rightarrow 1} + k_{4 \rightarrow 3} + k_{4 \rightarrow 5} + k_{4 \rightarrow 6}) & k_{5 \rightarrow 4} & k_{6 \rightarrow 4} \\ 0 & k_{2 \rightarrow 5} & 0 & k_{4 \rightarrow 5} & -(k_{5 \rightarrow 2} + k_{5 \rightarrow 4}) & 0 \\ 0 & k_{2 \rightarrow 6} & 0 & k_{4 \rightarrow 6} & 0 & -(k_{6 \rightarrow 2} + k_{6 \rightarrow 4}) \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{bmatrix} E1 \\ E2 \\ E3 \\ E4 \\ E5 \\ E6 \end{bmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

For definitions of the individual rate constants and their constituents (k_{QH2c3+} 's, k_{QH2} 's, and k_{SO} 's), see the original article. After solving for the state occupancy vector, the rate of ubiquinol oxidation is given as:

$$J_{C3}^{UQH_2} = E_{totC3} \left((k_{QH_2c^{3+}f}^{1 \rightarrow 2} - k_{QH_2r}^{4 \rightarrow 1})E1 + (k_{QH_2c^{3+}f}^{2 \rightarrow 6} - k_{QH_2c^{3+}r}^{1 \rightarrow 2} - k_{QH_2r}^{5 \rightarrow 2})E2 + k_{QH_2c^{3+}f}^{3 \rightarrow 4}E3 + (k_{QH_2c^{3+}f}^{4 \rightarrow 5} - k_{QH_2c^{3+}r}^{3 \rightarrow 4} - k_{QH_2f}^{4 \rightarrow 1})E4 \right. \\ \left. + (k_{QH_2f}^{5 \rightarrow 2} - k_{QH_2c^{3+}r}^{4 \rightarrow 5})E5 - k_{QH_2c^{3+}r}^{2 \rightarrow 6}E6 \right).$$

The rate of superoxide generation is given as:

$$J_{C3}^{SO} = E_{totC3} \left(-kSO_r^{2 \rightarrow 1}S1 + (kSO_f^{2 \rightarrow 1} - kSO_r^{6 \rightarrow 2})S2 - kSO_r^{4 \rightarrow 3}S3 + (kSO_f^{4 \rightarrow 3} - kSO_r^{5 \rightarrow 4})S4 + kSO_f^{5 \rightarrow 4}S5 + kSO_f^{6 \rightarrow 2}S6 \right).$$

And for numerical stability, the rate of cytochrome c reduction is coupled to the above rates such that:

$$J_{C3}^{c3} = 2J_{C3}^{UQH_2} - J_{C3}^{SO}.$$

Note that some equations are presented slightly differently than in the original article. See the Matlab codes for latest formulation. Below is an updated list of parameters.

Table S3.3. Ubiquinol-Cytochrome-c Oxidoreductase Parameters^a

Parameter	Definition	Value	Reference
Structural Parameters			
K_{stabo}	Q _o -site semiquinone stability constant	10^{-33} unitless	[14] ^{b,c}
K_{stabi}	Q _i -site semiquinone stability constant	$10^{-16.11}$ unitless	[15] ^c
pK_{ISPox1}	Oxidized Rieske ISP acidic pK	7.6	[16]
pK_{ISPox2}	Oxidized Rieske ISP alkaline pK	9.2	[16]
pK_{bLox}	Oxidized cytochrome b _L pK	5.9	[17]
pK_{bLred}	Reduced cytochrome b _L pK	7.9	[17]
pK_{bHox}	Oxidized cytochrome b _H pK	5.7	[17]
pK_{bHred}	Reduced cytochrome b _H pK	7.7	[17]
β	Fractional charge transfer coefficient	0.5 unitless	[17-20]
Kinetic and ROS Parameters			
E_{totC3}	Total Complex III content	1.58×10^{-3} mol/lmito	- ^d
k_{120}	State E1 to E2 intrinsic transition-rate constant	3.07×10^3 s ⁻¹	[3]
k_{230}	State E2 to E3 intrinsic transition-rate constant	3.31×10^7 s ⁻¹	[3]
k_{410}	State E4 to E1 intrinsic transition-rate constant	1.44×10^4 s ⁻¹	[3]
k_{340}	State E3 to E4 intrinsic transition-rate constant	3.07×10^3 s ⁻¹	- ^e
k_{260}	State E2 to E6 intrinsic transition-rate constant	3.07×10^3 s ⁻¹	- ^e
k_{640}	State E6 to E4 intrinsic transition-rate constant	3.31×10^7 s ⁻¹	- ^f
k_{450}	State E4 to E5 intrinsic transition-rate constant	3 s ⁻¹	- ^g

k_{520}	State E5 to E2 intrinsic transition-rate constant	$1.44 \times 10^4 \text{ s}^{-1}$	- ^h
k_{SO}	1st order superoxide production-rate constant	1600 s^{-1}	[14] ⁱ
K_{c3}	Cytochrome c^{3+} binding constant	$1.1 \text{ } \mu\text{M}$	[3]
K_{c2}	Cytochrome c^{2+} binding constant	$1.2 \text{ } \mu\text{M}$	[3]
$K_{QH_2}^o$	$Q_{10}H_2$ Q_o -site binding constant	0.8 mM	[21] ^j
K_Q^o	Q_{10} Q_o -site binding constant	0.5 mM	[21-23] ^k
K_Q^i	$Q_{10}H_2$ Q_i -site binding constant	1.0 mM	[24] ^l
$K_{QH_2}^i$	Q_{10} Q_i -site binding constant	2.4 mM	- ^m

^a Updated from [3].

^b Based on approximate K_{eq} for the superoxide reaction.

^c Extrapolated to pH 0 from given reference. At pH 7, $K_{stabo} = 10^{-9}$ and $K_{stabi} = 10^{-2.11}$.

^d Adjustable parameter.

^e Identical state transition as E1 to E2.

^f Identical state transition as E2 to E3.

^g This rate must be low to simulate physiological superoxide production rates.

^h Identical state transition as E4 to E1.

ⁱ Adjusted to within uncertainty range.

^j Based on the observation that maximum activity in the absence of a proton-motive force when the Qpool is 50% reduced.

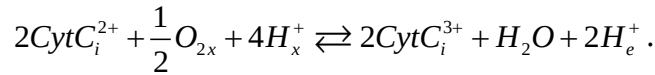
^k In the range given, and the Q_o -site affinity for QH_2 and Q are very similar.

^l Based on state-2 respiration rates of $30 \text{ nmol O}_2/\text{mg}/\text{min}$ when $\Delta\Psi = 200 \text{ mV}$ and $\Delta\text{pH} = 0$.

^m Defined via microscopic reversibility where $K_{QH_2}^i = K_{QH_2}^o {}^2K_Q^i K_{c3}^2 / K_Q^o / K_{c2}^2$.

Cytochrome-c oxidase: Complex IV

The biochemical equation for Complex IV is defined as



The equilibrium constant is defined as

$$K_{eq} = e^{-\left(\Delta_r G_{C4}^0 - 4\Delta\Psi\right)/RT} \left(\left[H^+ \right]_x^4 / \left[H^+ \right]_i^2 \right).$$

The rate expression used in the model is

$$J_{C4} = X_{C4} \left(\frac{1}{1 + K_{O_2} / [O_2]_x} \right) \left(\frac{[c^{2+}]_i^n}{[c^{2+}]_i^n + K_M^n (1 + \beta e^{2\Delta\Psi F/RT})} \right) \left(1 - \frac{[c^{3+}]_i^2}{[c^{2+}]_i^2 [O_2]_x^{1/2} K_{eq}^{1/2}} \right).$$

Table S3.4. Cytochrome-c Oxidase Parameters

Parameter	Definition	Value	Reference
X_{C4}	Complex IV activity	8.91×10^{-2} mol/lmito/s	- ^a
$\Delta_r G_{C4}^0$	Gibb's free energy of reaction	-202.16 kJ/mol	[5]
K_{O_2}	O ₂ binding constant	1×10^{-6} M	[1]
K_M	c ²⁺ binding constant	162 μM	[25]
n	Hill coefficient for c ²⁺	2	[25]
β	ΔΨ constant	6.6×10^{-6}	[25]

^a Adjustable parameter.

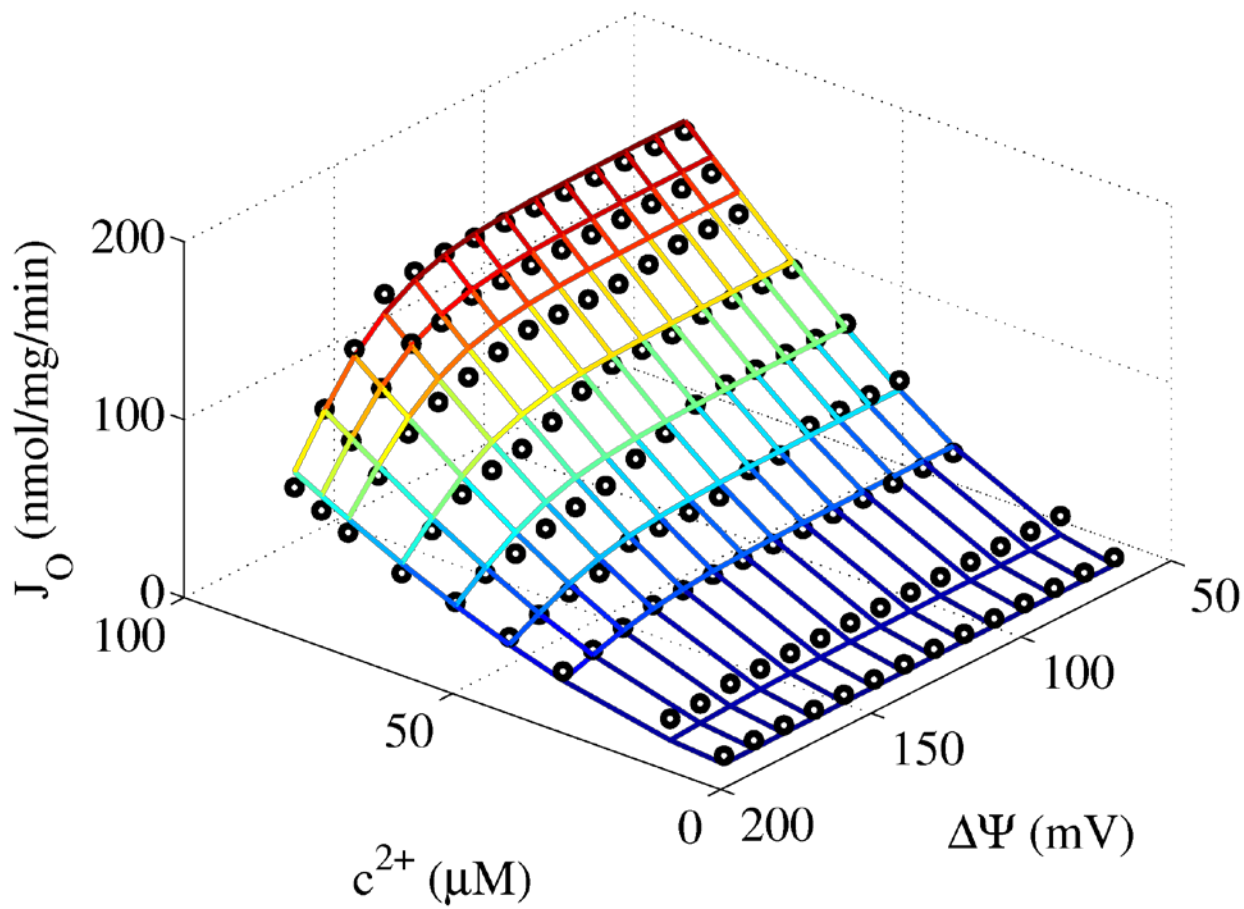


Figure S3.1. Flux through Complex IV as a function of c^{2+} and $\Delta\Psi$. The data from Murphy and Brand [25] was used to fit the K_M , n , and β parameters in the flux expression for Complex IV. The reported fraction of c^{2+} was converted to concentration using the measurements given in Estabrook and Holowinsky [26].

Proton leak

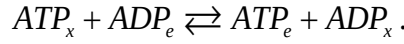
The electrophoretically driven proton uptake via leak pathways was updated to account for the highly non-linear nature of the leak across energy-transducing membranes [27]. The proton leak permeability, P_{Hleak} , is an adjustable parameter and set to 1.78×10^3 mol/s/lmito/M:

$$J_{Hleak} = P_{Hleak} \left([H^+]_e e^{\left(\frac{F\Delta\Psi}{2RT}\right)} - [H^+]_m e^{-\left(\frac{F\Delta\Psi}{2RT}\right)} \right).$$

S3B - Oxidative Phosphorylation Related Reactions Rates

Adenine nucleotide translocase (ANT)

The biochemical equation for ANT is defined as



The rate expression used in the model is

$$J_{ANT} = E_{ANT} \frac{k_2^{ANT} q \frac{[ATP^{4-}]_x [ADP^{3-}]_e}{K_o^D} - k_3^{ANT} \frac{[ATP^{4-}]_e [ADP^{3-}]_x}{K_o^T}}{\left(1 + \frac{[ATP^{4-}]_e}{K_o^T} + \frac{[ADP^{3-}]_e}{K_o^D}\right) ([ATP^{4-}]_x + q[ADP^{3-}]_x)},$$

where

$$k_2^{ANT} = k_2^{ANT,o} e^{(-3\alpha_1 - 4\alpha_2 + \alpha_3)F\Delta\Psi/RT},$$

$$k_3^{ANT} = k_3^{ANT,o} e^{(-4\alpha_1 - 3\alpha_2 + \alpha_3)F\Delta\Psi/RT},$$

$$K_o^D = K_o^{D,o} e^{3\delta_D RT/F\Delta\Psi},$$

$$K_o^T = K_o^{T,o} e^{4\delta_T RT/F\Delta\Psi},$$

and

$$q = \frac{k_3^{ANT} K_o^D}{k_2^{ANT} K_o^T} e^{F\Delta\Psi/RT}.$$

Table S3.5. Adenine Nucleotide Translocase Parameters^a

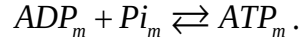
Parameter	Definition	Value	Reference
E_{ANT}	Total ANT content	0.141 mol/lmito	- ^b
$k_2^{ANT,o}$	Forward translocation rate	0.159 s ⁻¹	[5]
$k_3^{ANT,o}$	Reverse translocation rate	0.501 s ⁻¹	[5]
$K_o^{D,o}$	ADP binding constant	38.89 μM	[5]
$K_o^{T,o}$	ATP binding constant	56.05 μM	[5]
α_1	Translocation displacement constant	0.2829 unitless	[5]
α_2	Translocation displacement constant	-0.2086 unitless	[5]
α_3	Translocation displacement constant	0.2372 unitless	[5]
δ_T	ATP displacement binding constant	0.0167 unitless	[5]
δ_D	ADP displacement binding constant	0.0699 unitless	[5]

^a Updated from Metelkin et al. [28].

^b Adjustable parameter.

F₁F₀ ATP synthase

The biochemical equation for F₁F₀ ATP synthase is defined as



The equilibrium constant is defined as

$$K_{eq} = e^{-\left(\Delta_r G_{F_1F_0}^0 + 3F\Delta\Psi\right)/RT} \left(\frac{[H^+]_e^{nH}}{[H^+]_x^{nH-1}} \right) \left(\frac{P_{ATP}}{P_{ADP}P_{Pi}} \right).$$

The rate expression used in the model is

$$J_{F_1F_0} = X_{F_1F_0} ([ADP]_x [Pi]_x K_{eq} - [ATP]_x).$$

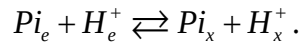
Table S3.6. F₁F₀ ATP synthase Parameters

Parameter	Definition	Value	Reference
$X_{F_1F_0}$	F ₁ F ₀ activity	812 mol/s/M/lmito	[1] ^a
$\Delta_r G_{F_1F_0}^0$	Gibb's free energy of reaction	-4.51 kJ/mol	[5]
nH	H ⁺ :ATP ratio	8/3	[29]

^a The activity was lowered from the cited value to reduced the stiffness of the system of DAEs. This resulted in negligible differences in state variable dynamics for the simulations that were used to identify this parameter value.

Inorganic phosphate carrier (PiC)

The biochemical equation for the PiC is defined as



The rate expression used in the model is

$$J_{PiC} = X_{PiC} \left(\frac{[Pi^-]_x [H^+]_x - [Pi^-]_e [H^+]_e}{[Pi^-]_e + k_{PiC}} \right).$$

Table S3.7 Inorganic Phosphate Parameters

Parameter	Definition	Value	Reference
X_{PiC}	PiC activity	3.34x10 ⁷ mol/s/M/lmito	[1]
k_{PiC}	Pi binding constant	1.61 mM	[1]

ATPase

The biochemical equation for an ATPase is defined as



The equilibrium constant is defined as

$$K_{eq} = e^{-\left(\Delta_r G_{ATPase}^0 / RT\right)} \left([H^+]_e^{-1}\right) \left(\frac{P_{ADP} P_{Pi}}{P_{ATP}}\right).$$

The rate expression used in the model is

$$J_{ATPase} = X_{ATPase} / \left(1 + [ADP]_e / K_{i,ADP}\right) \left(1 - [ADP]_e [Pi]_e / [ATP]_e / K_{eq}\right).$$

Table S3.8. ATPase Parameters

Parameter	Definition	Value	Reference
X_{ATPase}	ATPase activity	0 - 1 U/ml	^a
$\Delta_r G_{ATPase}^0$	Gibb's free energy of reaction	4.51 kJ/mol	[5]
$K_{i,ADP}$	ADP inhibition constant	2.41×10^{-4} M	^b

^a The value was based on the amount of ATPase added as given in the experimental companion paper. 1 U is defined as 1 μ mol/min of ATP hydrolyzed in the buffer.

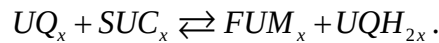
^b Adjustable parameter.

S3C – Ischemia/Reperfusion Reaction Rates

Note, the Complex II, adenylate kinase, creatine kinase, and external ATPase reactions shown below were only used for the ischemia/reperfusion simulations. In addition, the extra-mitochondrial water space to mitochondrial volume ratio, Vol_e , was set to 3 in accordance with measurements of cardiomyocyte density and composition data [6].

Succinate-ubiquinone oxidoreductase: Complex II

The biochemical equation for Complex II is defined as



The rate expression used in the model is

$$J_{C2} = X_{C2} \left(1 - [UQH_2]_x / [UQ]_x / K_{eq}^{C2}\right).$$

Table S3.9. Succinate-Ubiquinone Oxidoreductase Parameters

Parameter	Definition	Value	Reference
X_{C2}	Complex II activity	6.0×10^{-4} M/s/lmito	^a
K_{eq}^{C2}	Complex II equilibrium constant	1	-

^aThis value is set to give an approximate J_{CII}/J_{CI} ratio of 0.25.

Adenylate Kinase (AK)

The biochemical equation for AK is defined as



The equilibrium constant is defined as

$$K_{eq} = e^{-\left(\Delta_r G_{AK}^0 / RT\right)} \left(\frac{P_{ATP} P_{AMP}}{P_{ADP}^2} \right).$$

The rate expression used in the model is

$$J_{AK} = X_{AK} \left(1 - [ATP]_e [AMP]_e / [ADP]_e^2 / K_{eq} \right).$$

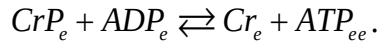
Table S3.10. AK Parameters

Parameter	Definition	Value	Reference
X_{AK}	AK activity	10^6 M/s	^a
$\Delta_r G_{AK}^0$	Gibb's free energy of reaction	2.38 kJ/mol	[5]

^aSet arbitrarily high to ensure equilibrium.

Creatine Kinase (CK)

The biochemical equation for CK is defined as



The equilibrium constant is defined as

$$K_{eq} = e^{-\left(\Delta_r G_{CK}^0 / RT\right)} [H^+]_e \left(\frac{P_{ATP}}{P_{ADP}} \right).$$

The rate expression used in the model is

$$J_{CK} = X_{CK} \left(1 - [ATP]_e [AMP]_e / [ADP]_e^2 / K_{eq} \right).$$

Table S3.11. AK Parameters

Parameter	Definition	Value	Reference
X_{CK}	CK activity	10^6 M/s	^a
$\Delta_r G_{CK}^0$	Gibb's free energy of reaction	-7.91 kJ/mol	[5]

^aSet arbitrarily high to ensure equilibrium.

ATPase

The biochemical equation for an ATPase is defined as



The rate expression used in the model is

$$J_{ATPase} = \frac{X_{ATPase}}{1 + R \frac{[ADP]_e [Pi]_e}{[ATP]_e}} .$$

Table S3.12. ATPase Parameters

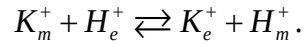
Parameter	Definition	Value	Reference
X_{ATPase}	ATPase activity	2.0×10^{-3} M/s	^a
R	Critical value of mass-action ratio of ATP hydrolysis potential	6.58×10^4 M ⁻¹	[5]

^aSet to achieve an intermedial level of respiration.

S3D - Cation Related Reaction Rates

Potassium-hydrogen exchanger

The biochemical equation for KHE is defined as



The rate expression used in the model is

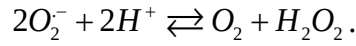
$$J_{KHE} = X_{KHE} \left([K^+]_m [H^+]_e - [K^+]_e [H^+]_m \right)$$

where X_{KHE} is set to 4.76×10^6 mol/s/M/lmito as in [1].

S4E – ROS Scavenging Related Reactions Rates

Manganese Superoxide Dismutase (MnSOD)

The biochemical equation for MnSOD is defined as



The rate expression used in the model is

$$J_{SODx} = X_{SOD}[O_2^-]_x$$

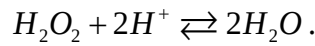
and

$$J_{SODe} = X_{SOD}[O_2^-]_e$$

where the subscripts x and e denote matrix and extra-mitochondrial, respectively, X_{SOD} is set to $2.0 \times 10^4 \text{ s}^{-1}$ based on $10 \text{ }\mu\text{M}$ [MnSOD] and a second order rate constant of $2.0 \times 10^9 \text{ M}^{-1}\text{s}^{-1}$ as given in [30, 31]. For simplicity, the extra-mitochondrial activity was set equal to the mitochondrial activity.

H₂O₂ Reduction via GSH and Trx Pathways

The biochemical equation for H₂O₂ scavenging is defined as



The rate expression used in the model is

$$J_{H_2O_2} = X_{H_2O_2}[H_2O_2]_x$$

where $X_{H_2O_2}$ is set to $1.32 \times 10^4 \text{ s}^{-1}$ based on $60 \text{ }\mu\text{M}$ [Prx3] and a second order rate constant of $2.0 \times 10^7 \text{ M}^{-1}\text{s}^{-1}$ and $2 \text{ }\mu\text{M}$ [Gpx1] and a second order rate constant of $6.0 \times 10^7 \text{ M}^{-1}\text{s}^{-1}$ as given in [30, 31].

The expression for H₂O₂ permeation through the mitochondrial membrane is

$$J_{H_2O_2 \text{ perm}} = k_{\text{perm}} ([H_2O_2]_x - [H_2O_2]_e)$$

where k_{perm} is set to 333 s^{-1} based on a membrane permeability coefficient of $2 \text{ }\mu\text{m/s}$ [32] and a membrane thickness of 6 nm .

S4: Simulation Protocols, Sensitivity Analysis, and Control Coefficients

For specific details of the simulation protocols, see the available Matlab codes. In brief, for each Pi condition, 1 mM or 5mM, the model was initialized for 60 s of simulation time. Then the external ATPase (Apyrase) rate was adjusted to the rate reported in the experimental companion paper, and the model was simulated for 5 minutes. To remain faithful to the experimental protocol, the state variable trajectories were compared to the experimental data after 3 minutes of simulation time. As such, some model state variables may have not reached their steady state values while others are quite close. The rate of approach to steady state is set by the slowest component in the system. In this case, it is the overall mitochondrial ATP synthesis and transport rate that sets the rate.

For the sensitivity analysis, the complex method [33] was used to approximate the first order derivatives of the model outputs (e.g. NADH, $\Delta\Psi$, Cyt c^{2+} , Buffer ADP, and V_{O_2}) with respect to the parameter values after 3 minutes of simulation time. Then, for each parameter, all the non-zero sensitivity coefficients were averaged and are reported in Table 1 of the main article.

For the calculation of the control coefficients, the method outlined in Kascr and Burns was used [34]. Specifically, the sensitivities for all the independent fluxes and state variables were taken with respect to the parameter values and the state variables using the complex method at the simulation time corresponding to the steady state in the experimental protocol (approximately 3 minutes). The normalized flux control coefficients for a given flux taken with respect to a given reaction are computed by taking the product of the derivative of that flux with respect to the enzyme concentration and normalized by multiplying the enzyme concentration and dividing by the flux as detailed in Kascr and Burns.

S5: Steady State Analysis

Representative simulations for the model dynamics for the simulated experiments are given below in Figure S5.1.

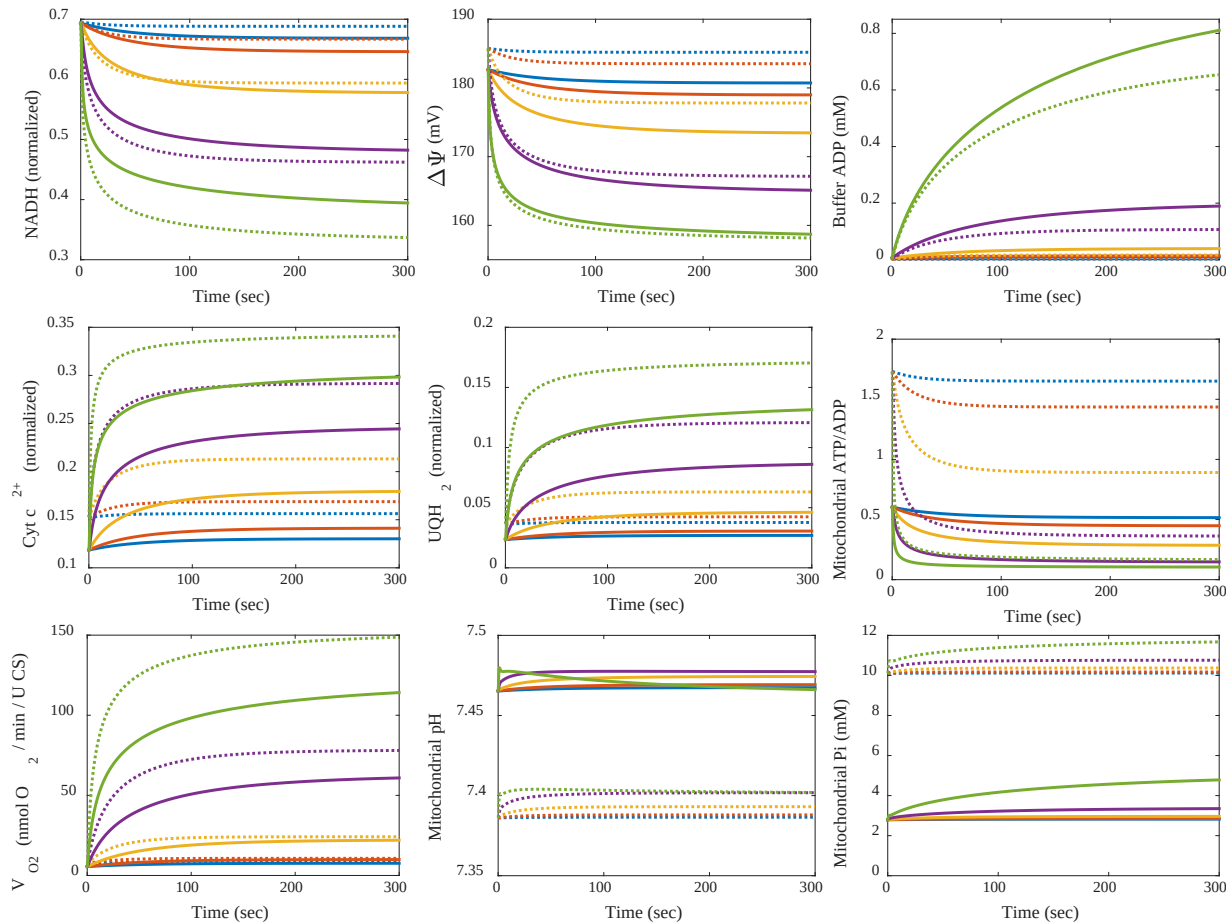


Figure S5.1. Model dynamics of key state variables for the simulation protocol used to reproduce the experimental data in the companion paper. For this simulation, the model was simulated for 5 minutes under the experimental conditions described in section S4 of this document. The ATPase rate was set to 0 (blue), 0.25 (orange), 0.5 (yellow), 0.75 (purple), and 1 (green) U/ml for both the 1 mM Pi case (solid line) and the 5 mM Pi case (dashed line).

For the steady state analysis, the model was simulated until the absolute values of the state variable derivatives were less than 10^{-10} for both the 1 mM and 5 mM Pi condition for all ATPase rates. The fixed point was perturbed by $\pm 1\%$ to generate a new initial condition. For conserved moieties such as mitochondrial adenine nucleotides, nicotinamide adenine dinucleotides, ubiquinone, cytochrome c, and cytosolic phosphate, the conservation was strictly enforced. That said, $[O_2]$ was not fixed and, as such, shifted the concentration of free radicals to new steady states. The perturbation resulted in a displacement of varying distances or δ 's. The model was then simulated to steady state, and a new fixed point was established. This resulted in displacements (from the original fixed point) of varying distances or ϵ 's. The results of 100 random perturbations for all ATPase rates are given in Figure S5.2. For the 1 mM Pi case, the δ 's range from approximately 0.015 to 0.03, and the corresponding ϵ 's range from approximately 0.01 to 0.12. For the 5 mM Pi case, the δ 's have an identical range as in the 1 mM Pi case, but the corresponding ϵ 's range from approximately 0.01 to 0.04. Thus, the fixed point

is Lyapunov stable since for every $\epsilon > 0$, there exists a δ such that $\|\mathbf{x}(t) - \mathbf{x}^*\| < \epsilon$ for $t \geq 0$ and $\|\mathbf{x}(0) - \mathbf{x}^*\| < \delta$. In other words, trajectories that start within δ of $\mathbf{x}^*$ remain within ϵ of $\mathbf{x}^*$ for all time.

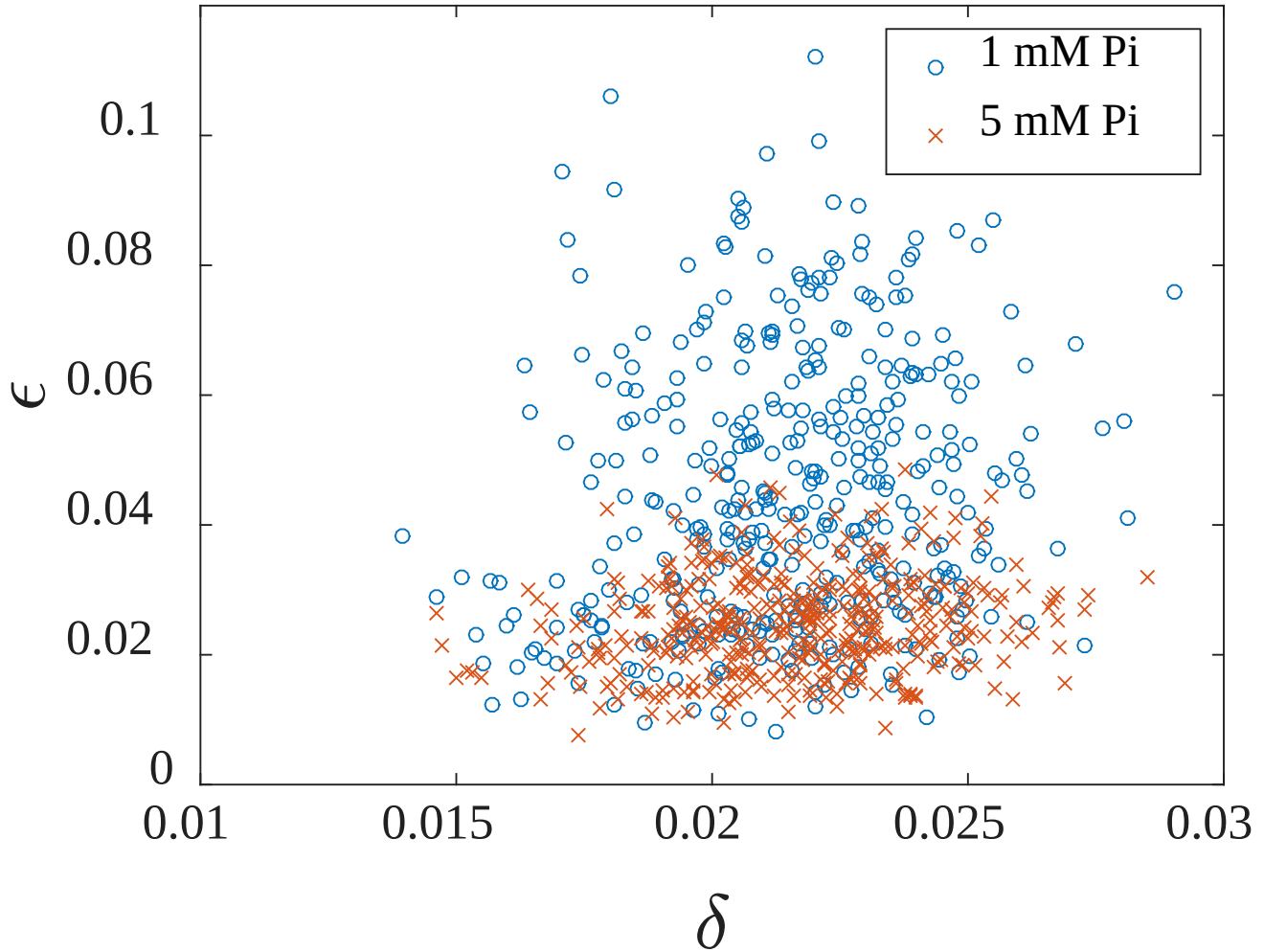


Figure S5.2. Perturbation analysis of model fixed points at 1 mM Pi and 5 mM Pi for a given external ATPase rate. For details, see text.

S6: Model Code Details

[Simulate_Vinnakota2015.m](#) – script used to call the necessary functions to simulate the model and reproduce the plots for the Vinnakota data set as given in the main article.

[Simulate_FCCP_Titration.m](#) – script used to call the necessary functions to simulate the model and reproduce the plots for the FCCP titration as given in the main article.

[Simulate_IR.m](#) – script used to call the necessary functions to simulate the model and reproduce the plots for the ischemia/reperfusion conditions as given in the main article.

[DAEs.m](#) – a function that defines the right hand side of the system of differential algebraic equations that comprise the model.

[Model_IC.m](#) – a function that defines the model initial conditions.

[dGpKcorr.m](#) – a function that corrects the Gibb’s free energies of formation and pK’s for ionic strength and temperature.

[herrorbar.m](#) – a function by Jos van der Geest that plots horizontal error bars and downloaded from the Matlab File Exchange server.

[data.mat](#) – cell array of experimental data.

[params.mat](#) – structure containing model parameters.

References

1. Beard, D.A., *A biophysical model of the mitochondrial respiratory system and oxidative phosphorylation*. PLoS Comput Biol, 2005. **1**(4): p. e36.
2. Bazil, J.N., et al., *Determining the origins of superoxide and hydrogen peroxide in the mammalian NADH:ubiquinone oxidoreductase*. Free Radic Biol Med, 2014. **77**: p. 121-9.
3. Bazil, J.N., et al., *Analysis of the kinetics and bistability of ubiquinol:cytochrome c oxidoreductase*. Biophys J, 2013. **105**(2): p. 343-55.
4. Li, X., et al., *A database of thermodynamic properties of the reactions of glycolysis, the tricarboxylic acid cycle, and the pentose phosphate pathway*. Database (Oxford), 2011. **2011**: p. bar005.
5. Wu, F., et al., *Phosphate metabolite concentrations and ATP hydrolysis potential in normal and ischaemic hearts*. J Physiol, 2008. **586**(Pt 17): p. 4193-208.
6. Vinnakota, K.C. and J.B. Bassingthwaighe, *Myocardial density and composition: a basis for calculating intracellular metabolite concentrations*. Am J Physiol Heart Circ Physiol, 2004. **286**(5): p. H1742-9.
7. Alberty, R.A., *Thermodynamics of biochemical reactions*. 2003, Hoboken, N.J.: Wiley-Interscience. ix, 397 p.
8. Petlicki, J. and T.G.M. van de Ven, *The equilibrium between the oxidation of hydrogen peroxide by oxygen and the dismutation of peroxy or superoxide radicals in aqueous solutions in contact with oxygen*. Journal of the Chemical Society-Faraday Transactions, 1998. **94**(18): p. 2763-2767.
9. Vinnakota, K.C., et al., *Multiple ion binding equilibria, reaction kinetics, and thermodynamics in dynamic models of biochemical pathways*. Methods Enzymol, 2009. **454**: p. 29-68.
10. Wu, F., et al., *Computer modeling of mitochondrial tricarboxylic acid cycle, oxidative phosphorylation, metabolite transport, and electrophysiology*. J Biol Chem, 2007. **282**(34): p. 24525-37.
11. Jung, D.W., et al., *On the relationship between matrix free Mg^{2+} concentration and total Mg^{2+} in heart mitochondria*. Biochim Biophys Acta, 1997. **1320**(3): p. 310-20.
12. Sled, V.D., et al., *Thermodynamic analysis of flavin in mitochondrial NADH:ubiquinone oxidoreductase (complex I)*. Biochemistry, 1994. **33**(33): p. 10069-75.
13. Ingledew, W.J. and T. Ohnishi, *An analysis of some thermodynamic properties of iron-sulphur centres in site I of mitochondria*. Biochem J, 1980. **186**(1): p. 111-7.
14. Buettner, G.R., et al., *A new paradigm: manganese superoxide dismutase influences the production of H_2O_2 in cells and thereby their biological state*. Free Radic Biol Med, 2006. **41**(8): p. 1338-50.
15. Ohnishi, T. and B.L. Trumpower, *Differential effects of antimycin on ubiquinone bound in different environments in isolated succinate . cytochrome c reductase complex*. J Biol Chem, 1980. **255**(8): p. 3278-84.
16. Link, T.A., et al., *Determination of the redox properties of the Rieske $[2Fe-2S]$ cluster of bovine heart bc1 complex by direct electrochemistry of a water-soluble fragment*. Eur J Biochem, 1992. **208**(3): p. 685-91.
17. Rich, P.R., et al., *Inhibitor effects on redox-linked protonations of the b haems of the mitochondrial bc1 complex*. Biochim Biophys Acta, 1990. **1018**(1): p. 29-40.
18. Glaser, E.G., S.W. Meinhardt, and A.R. Crofts, *Reduction of cytochrome b-561 through the antimycin-sensitive site of the ubiquinol-cytochrome c2 oxidoreductase complex of Rhodopseudomonas sphaeroides*. FEBS Lett, 1984. **178**(2): p. 336-42.
19. Glaser, E.G. and A.R. Crofts, *A new electrogenic step in the ubiquinol:cytochrome c2 oxidoreductase complex of Rhodopseudomonas sphaeroides*. Biochim Biophys Acta, 1984. **766**(2): p. 322-33.
20. Robertson, D.E. and P.L. Dutton, *The nature and magnitude of the charge-separation reactions of ubiquinol cytochrome c2 oxidoreductase*. Biochim Biophys Acta, 1988. **935**(3): p. 273-91.
21. Crofts, A.R., *The cytochrome bc1 complex: function in the context of structure*. Annu Rev Physiol, 2004. **66**: p. 689-733.
22. Ding, H., et al., *Ubiquinone pair in the Qo site central to the primary energy conversion reactions of cytochrome bc1 complex*. Biochemistry, 1995. **34**(49): p. 15979-96.

23. Ding, H., et al., *Cytochrome bc₁ complex [2Fe-2S] cluster and its interaction with ubiquinone and ubihydroquinone at the Qo site: a double-occupancy Qo site model*. *Biochemistry*, 1992. **31**(12): p. 3144-58.
24. Bose, S., et al., *Metabolic network control of oxidative phosphorylation: multiple roles of inorganic phosphate*. *J Biol Chem*, 2003. **278**(40): p. 39155-65.
25. Murphy, M.P. and M.D. Brand, *The control of electron flux through cytochrome oxidase*. *Biochem J*, 1987. **243**(2): p. 499-505.
26. Estabrook, R.W. and A. Holowinsky, *Studies on the content and organization of the respiratory enzymes of mitochondria*. *J Biophys Biochem Cytol*, 1961. **9**: p. 19-28.
27. Garlid, K.D., et al., *On the mechanism by which dicyclohexylcarbodiimide and quinine inhibit K⁺ transport in rat liver mitochondria*. *J Biol Chem*, 1986. **261**(4): p. 1529-35.
28. Metelkin, E., I. Goryanin, and O. Demin, *Mathematical modeling of mitochondrial adenine nucleotide translocase*. *Biophys J*, 2006. **90**(2): p. 423-32.
29. Nicholls, D.G. and S.J. Ferguson, *Bioenergetics*. Fourth edition / ed. 2013, Amsterdam: Academic Press, Elsevier. xiv, 419 pages.
30. Murphy, M.P., *Mitochondrial thiols in antioxidant protection and redox signaling: distinct roles for glutathionylation and other thiol modifications*. *Antioxid Redox Signal*, 2012. **16**(6): p. 476-95.
31. Murphy, M.P., *How mitochondria produce reactive oxygen species*. *Biochem J*, 2009. **417**(1): p. 1-13.
32. Bienert, G.P., J.K. Schjoerring, and T.P. Jahn, *Membrane transport of hydrogen peroxide*. *Biochim Biophys Acta*, 2006. **1758**(8): p. 994-1003.
33. Squire, W. and G. Trapp, *Using complex variables to estimate derivatives of real functions*. *Siam Review*, 1998. **40**(1): p. 110-112.
34. Kacser, H. and J.A. Burns, *The control of flux*. *Symp Soc Exp Biol*, 1973. **27**: p. 65-104.